



Ahsan Farabi

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LinkedIn — GitHub — Google Scholar — Portfolio

RESEARCH INTERESTS

Large Language Models (LLMs), Multimodal Learning, Efficient and Sustainable AI, Human-Computer Interaction

EDUCATION

United International University, Dhaka, Bangladesh
B.Sc. in Computer Science & Engineering
Major: Data Science

2022 – 2026
CGPA: 3.88/4.00
Major CGPA: 4.00/4.00

Uttara Model College, Dhaka, Bangladesh
Higher Secondary Certificate (HSC), Group: Science

2018 – 2020
GPA: 5.00/5.00

Uttara Model School, Dhaka, Bangladesh
Secondary School Certificate (SSC), Group: Science

2016 – 2018
GPA: 5.00/5.00

TEACHING EXPERIENCE

Instructor

UIU Computer Club

Hybrid
July 2026 – Present

Workshop: From Idea to Deployment: A hands on ML Journey

- Conducted interactive, hands-on sessions on practical machine learning workflows.
- Demonstrated an agentic workflow to build ML projects from scratch using prompt engineering.
- Provided research directions and guidance for ML project development.

Undergraduate Teaching Assistant

United International University

Contractual
June 2024 – October 2025

Courses: Object-Oriented Programming Lab, Database Management Systems Lab

- Assisted in delivering laboratory sessions, technical demonstrations, and weekly consultation hours for 100 undergraduate students across two laboratory courses.
- Designed supplementary learning materials, programming exercises, and exam preparation resources to reinforce core concepts and improve student learning.
- Mentored student project teams throughout the semester; supervised a team that won the *Champion* award at the UIU CSE Project Show.

RESEARCH EXPERIENCE

GreenSwap: Verb Choice or Output Length? A Controlled Decomposition of Prompt-Level Energy Savings in LLM Inference

Accepted in Main Conference, Empirical Methods in Natural Language Processing (EMNLP 2026)

[code]

Supervised By: Shahrirar Mahmud

- Tested whether changing prompt wording or limiting response length saves more energy during LLM inference, across 21,600 test runs.
- Found that telling the model to keep answers short saves more energy and gives better quality than swapping words in the prompt.

EcoDetect: Lightweight Real-Time Recyclable Waste Detection Using YOLOv11n Trained on LLM-Guided Synthetic Data

Published, International Conference on Computer and Information Technology (ICCIT 2025)

[code] [paper]

Supervised By: Md. Tanvir Raihan

- Used AI-generated synthetic images to train a lightweight waste detection model, avoiding the need for manual data collection.
- Achieved 78.9% precision and 81.1% mAP@0.5 on real-world test images the model had never seen before.

WADE: A Waste Annotation Reasoning-Chain Dataset and Distillation Framework for Low-Resource Vision-Language Models

Submitted to *Engineering Applications of Artificial Intelligence (EAAI)*, Elsevier — Undergraduate Thesis

[code]

Supervised By: Dr. Muhammad Nomani Kabir

- Developed a multimodal dataset comprising 2,167 in-situ images of Bangladeshi water bodies with 13,608 manually annotated waste instances and reasoning chains using a custom annotation platform.
- Conducted a comprehensive benchmark of six vision-language models under zero-shot, few-shot, and reasoning-based prompting, identifying perception—not reasoning—as the primary limitation in aquatic waste understanding.
- Fine-tuned an open-source vision-language model using QLoRA, improving domain-specific perception while reducing hallucinations in complex aquatic environments.

SELECTED PROJECTS

Orchestra (Local AI Coding Agent) [github] [download] 2026

Technology: Python, Ollama, TUI

Built a privacy-first local AI coding agent with autonomous tool-calling, multi-session context, web search, and permission-gated safety controls, validated through structured acceptance testing.

Wiselearn (Python Library) [github] [download] 2026

Technology: Python, PyTorch, HuggingFace

Built an educational ML library for streamlined dataset preparation, model training, and explainability.

Dispatch (API Testing Tool) [github] [live] 2026

Technology: JavaScript, Next.js, Tailwind CSS, Zustand

Built a visual, node-based API testing platform with workflow automation, authentication, and endpoint validation.

EcoRover (Waste Sorting Robot) [github] [video] 2025

Technology: Python, OpenCV, YOLOv8n, Raspberry Pi 5, Arduino, Servo Motor Control

Built a 4-DOF robotic system that autonomously classifies and sorts recyclable waste in real time using computer vision.

UIU Platformer (Onboarding Game) [github] [video] 2024

Technology: Java, JavaFX

Built a 2D onboarding simulation game for incoming students and evaluated its usability through testing with 20+ participants.

AWARDS & CERTIFICATIONS

Finalist - National Blockchain Olympiad Bangladesh 2025

On-Campus Champion - Hult Prize UIU (Social Entrepreneurship Competition) 2026

1st Runner-Up - UIU CSE Project Show (DBMS Poster Award) 2023

4th Runner-Up - UIU CSE Project Show (Database Management Systems) 2023

2nd Runner-Up - UIU CSE Project Show (System Analysis & Design) 2024

2nd Runner-Up - UIU CSE Project Show (Advanced Object-Oriented Programming) 2024

4th Runner-Up - UIU CSE Project Show (Software Engineering) 2025

Merit Scholarship - United International University 2022 – 2026

Machine Learning Specialization - Stanford University [certificate] 2024

Git & Github - Google [certificate] 2024

SKILLS

Programming Languages: C, C++, Java, Python

Scripting Languages: BASH, HTML, PHP, CSS, JavaScript, SQL, LaTeX

AI & Machine Learning: NumPy, Pandas, Matplotlib, Scikit-Learn, PyTorch, Data Annotations, Agentic Workflow

Software Development: SDLC, Next.js, Tailwind CSS, Supabase, Vercel, Postman, Android Studio, Google Cloud

Version Control & DevOps: Git, GitHub, Linux, CI/CD

Hardware: Raspberry Pi, ESP32, Arduino

Tools: MS Word, PowerPoint, Excel, Jira, Canva, Miro, LucidSpark

Soft Skills: Communication, Leadership, Teaching, Design Thinking, Public Speaking, Curriculum Design

REFERENCES

Dr. Muhammad Nomani Kabir

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